CSE 594 Assignment 2

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For my task, I will have the task performers identify logical fallacies across a series of statements. On a custom web interface, participants will be presented with statements from the Hugging Face fallacy dataset https://huggingface.co/datasets/kuwrom/fallacy. They will be presented specifically with the "task" set. For each statement, task performers will be given 14 options for each type of fallacy that could apply to that statement, and performers must select which of these options is most applicable. Performers will be given a brief description of each fallacy option via web popup before the task, so they understand what they need to identify. AI assistance will be provided as an AI model answers each question alongside the user and provides a suggestion in the corner of their webpage. The user can then choose whether or not to use the model's suggestion for their own answer.

For this assignment, I trained my own model using DistilBERT on 5000 randomly selected statements within the "training" set from the fallacy dataset provided by Hugging Face. I also used 1000 randomly selected statements within the "validation" set to validate the model during training. I used Colab to train and download the model which I then ran on my personal laptop. The general process of training and running this model is illustrated below.



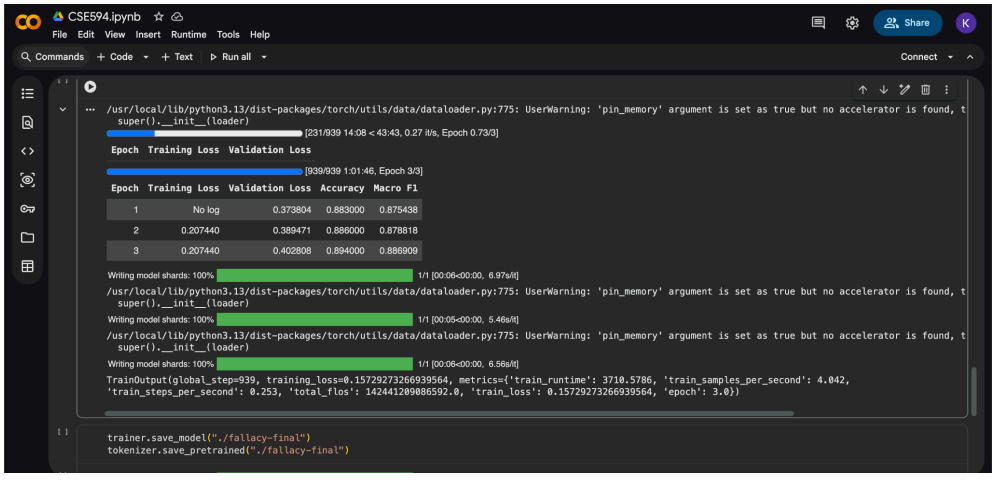
```
train_data = dataset["train"].shuffle(seed=42).select(range(5000))
validation_data = dataset["validation"].shuffle(seed=42).select(range(1000))

label_names = dataset["train"].features["label"].names

print(label_names)
```

```
['ad_hominem', 'ad_populum', 'appeal_to_authority', 'appeal_to_ignorance', 'cherry_picking', 'circular_reasoning', 'equivocation', 'false_causalit
```

Colab code to grab training and validation data for the model



CSE594.ipynb

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```
/usr/local/lib/python3.13/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, t
  super().__init__(loader)
```

[231/939 14:08 < 43:43, 0.27 it/s, Epoch 0.73/3]

Epoch	Training Loss	Validation Loss

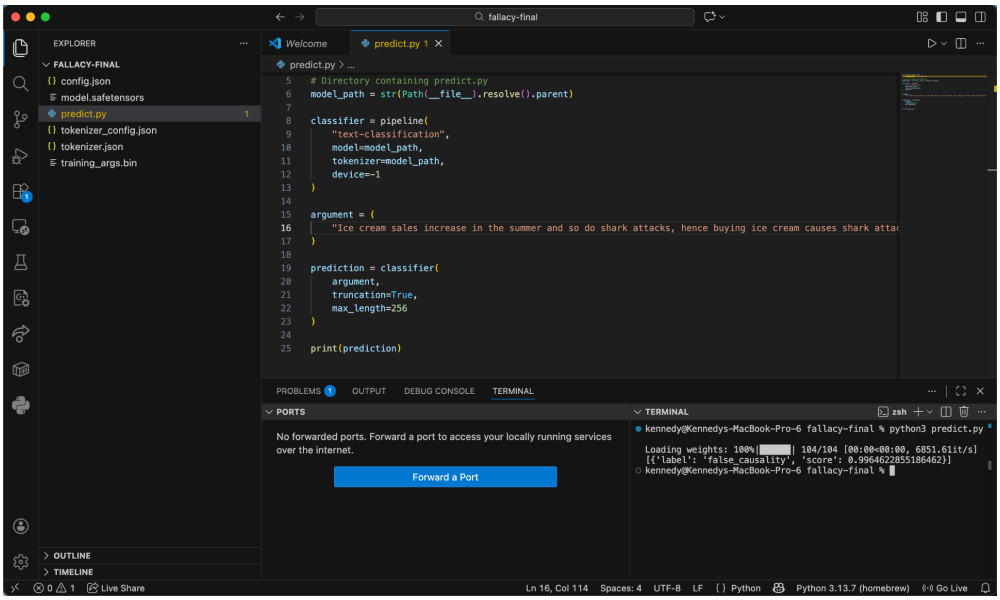
[939/939 1:01:46, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	Macro F1
1	No log	0.373804	0.883000	0.875438
2	0.207440	0.389471	0.886000	0.878818
3	0.207440	0.402808	0.894000	0.886909

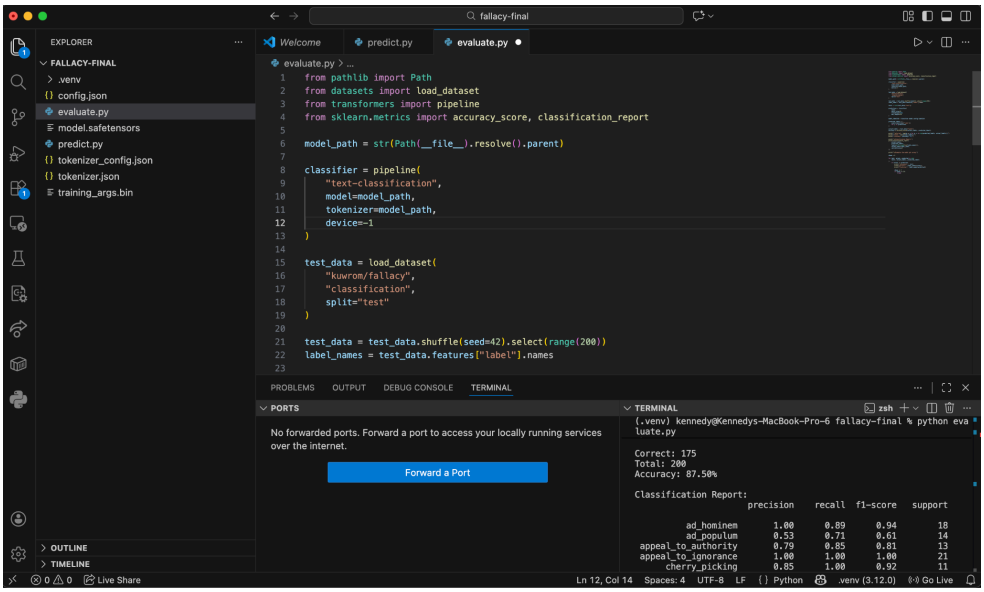
```
Writing model shards: 100% 1/1 [00:06<00:00, 6.97s/it]
/usr/local/lib/python3.13/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, t
  super().__init__(loader)
Writing model shards: 100% 1/1 [00:05<00:00, 5.46s/it]
/usr/local/lib/python3.13/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, t
  super().__init__(loader)
Writing model shards: 100% 1/1 [00:06<00:00, 6.56s/it]
TrainOutput(global_step=939, training_loss=0.15729273266939564, metrics={'train_runtime': 3710.5786, 'train_samples_per_second': 4.042,
'train_steps_per_second': 0.253, 'total_flos': 142441209086592.0, 'train_loss': 0.15729273266939564, 'epoch': 3.0})
```

```
trainer.save_model("./fallacy-final")
tokenizer.save_pretrained("./fallacy-final")
```

Training DistilBERT in Colab



Model running on one statement (full predict.py code in zip file)



Evaluating model on 200 statements (full evaluate.py code in zip file)

I believe that AI assistance will help with this task because logical fallacies are not always obvious. Human bias may influence fallacy identification: for example, more weight may be given to fallacies that participants are most familiar with. This is where an AI assistant could help navigate a human user in the right direction, giving them another option to consider before making their final decision.

I chose to evaluate my model based on accuracy, precision, and recall. Accuracy measures overall classification correctness, while precision and recall help identify whether

specific fallacies are frequently predicted incorrectly or overlooked. F1-score combines precision and recall, providing a balanced measure of classification performance. This allowed me to identify which logical fallacies had the highest error rates, and whether that was due to false positives or false negatives. I calculated the evaluation metrics using the standard formulas for precision, recall, and F1-score in Google Sheets.

It is also important to note that the model is being tested on ground-truth labels, which may not be a definitive source of truth if multiple labels apply. This is one limitation with this evaluation method. My model had an accuracy rating of 87.50% when run on the test set of 200 randomly selected statements. The 200 study trials were selected from the dataset's separate test split and were not used during training or validation. The model struggled most with the bandwagon fallacy, confusing it frequently with ad populum. Ad populum is the general fallacy that something is true because a large number of people believe it. The bandwagon differs slightly in that it argues that one should take a specific action because many people participate in it. This is the kind of small nuance that a model may struggle with, regardless of the training set. Outside of ad populum and bandwagon, the model also struggled a little with appeal to authority and red herring fallacies, with F1-scores of 0.81 and 0.86, respectively.

Fallacy Classification Report

Class	Precision	Recall	F1-score
ad_hominem	1.00	0.89	0.94
ad_populum	0.53	0.71	0.61
appeal_to_authority	0.79	0.85	0.81
appeal_to_ignorance	1.00	1.00	1.00
cherry_picking	0.85	1.00	0.92
circular_reasoning	1.00	0.85	0.92
equivocation	0.80	1.00	0.89
false_causality	1.00	1.00	1.00
false_dilemma	0.92	0.92	0.92
hasty_generalization	1.00	1.00	1.00
loaded_question	1.00	1.00	1.00
red_herring	0.90	0.82	0.86
slippery_slope	1.00	1.00	1.00
the_bandwagon	0.57	0.40	0.47
Accuracy			**0.88**

Macro avg	**0.88**	**0.89**	**0.88**
Weighted avg	**0.88**	**0.88**	**0.87**
Correct	175.00		
Total	200.00		
Accuracy	0.88		

Fallacy classification report for trained DistilBERT model (included in zip file)

Error pattern	Number of errors	Percent of all errors
Bandwagon → Ad populum	8	32%
Ad populum → Bandwagon	4	16%
Bandwagon → Appeal to authority	3	12%
Appeal to authority → Bandwagon	1	4%
Other classification errors	9	36%
Total	**25**	**100%**
Total study trials	**200**	
Correct classifications	175	
Incorrect classifications	25	
Accuracy	87.50%	

Error distribution chart for trained DistilBERT model (included in zip file)

Of the 200 trials evaluated, the model incorrectly classified 25 statements. The most frequent error pattern was confusion between the bandwagon and ad populum fallacies, accounting for 48% of all errors. An additional 16% of errors involved confusion between the bandwagon and appeal to authority fallacies. Together, these errors accounted for 64% of all misclassifications, suggesting that the model struggles most with distinguishing between conceptually similar fallacies. These mistakes present an opportunity for human-AI complementarity. Human performers may be better able to distinguish between fallacies by considering the context and intended meaning of each statement. However, some of these distinctions may also be difficult for humans, and incorrect AI suggestions could lead participants toward incorrect answers. This will be an important consideration when evaluating human-AI performance in Assignment 3.

I did not make any iterations to the model because, despite achieving an overall accuracy of 87.50%, it demonstrated identifiable weaknesses when distinguishing between similar logical

fallacies. These weaknesses may provide opportunities for human task performers to improve upon the model's predictions. However, whether humans can consistently recognize and correct these errors remains to be determined in Assignment 3.

My prediction is that AI-only performance will outperform human-only performance in classification accuracy, due to biases in concept familiarity. However, I also predict that AI-human performance will outperform both groups, since the AI's performance on its own is good, but has room for improvement. A human may be better at making the distinction between similar categories like ad populum and the bandwagon, while being able to rely on the AI if needed. Overall, my ranking for each condition would be:

- AI-human performance (highest accuracy)
- AI-only performance
- Human-only performance (lowest accuracy)